O-Drift: Observability-Aware Off-Policy Evaluation in Evolving Recommender Systems

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Abstract. Off-policy evaluation for the current period must often rely on logs collected over many earlier periods, with only a small sample from the period itself. Common remedies (recency windows, decay, forecasting) treat every change alike and discard data that is still informative. We argue that the useful question is not how recent a period is but which mechanism changed: a change in the context distribution or in the logging policy can be corrected by reweighting, while only a change in the conditional reward biases a reweighted estimate. We propose O-Drift, which segments the history with a per-action residual test that is invariant to context and logger drift, fits joint reward models across segments, reuses older segments with a drift correction, and pools the segment estimates at minimum variance with a random-effects interval. In a pre-registered evaluation with a one-shot test split, O-Drift outperforms a cross-domain estimator that groups periods by mean reward in all 15 reward-drift settings, and replacing its residual test by mean-reward grouping causes spurious splits under context and logger drift. On semi-synthetic Open Bandit data, however, drift-corrected reuse is strongly biased; we report this failure and show why validation on the small current-period sample cannot repair it.

Keywords: Off-policy evaluation · Non-stationarity · Concept drift · Recommender systems

1 Introduction

Recommender systems decide whether to deploy a policy by estimating its value from logs of earlier policies, which off-policy evaluation (OPE) allows when the logs record propensities [10,8]. The logs span many weeks while users change, loggers are replaced and click behaviour moves; what is needed is the value *now*, in a target period with only a small sample.

Pooling all history gives a low-variance estimate that is biased whenever the environment has changed; using only the small target sample (the *anchor*) is unbiased but noisy. Existing remedies interpolate between the two along the time axis: sliding windows and moving averages [11], exponential decay, forecasting the policy value as a time series [20,3], and regression-assisted estimators that reuse a model fitted on recent data [13]. All of them assume that older data is less useful because it is older.

This assumption ignores what changed. A logged record is generated by three mechanisms: the context distribution $P_t(x)$, the logging policy $\mu_t(a \mid x)$ and the conditional reward $q_t(x, a) = \mathbb{E}[R \mid x, a, t]$. The first two are nuisances that standard tools already correct: importance weights handle the logger [1], and density-ratio transport handles the contexts [21]. Only a change in q_t biases a correctly reweighted estimate. An old period whose users looked different but clicked in the same way is as informative as yesterday; a recent period after a change in click behaviour is not. Recency is a proxy for the mechanism that matters, and a poor one when contexts or loggers drift more often than rewards.

We turn this observation into an estimator, O-Drift (off-policy evaluation under drift). It tests for drift in the conditional reward directly, with a per-action residual statistic computed from one pooled reward model, and segments the history by binary segmentation. Under correct specification the test keeps its nominal level under any context or logger drift (Proposition 1), whereas statistics built from raw rewards, such as the mean-reward similarity used by the cross-domain estimator COPE [14], do not. Within each segment O-Drift pools windows with balanced multiple-logger weights; across segments it fits joint reward models with a shared base and per-segment offsets, reuses older segments after correcting their reward offset, and combines the segment estimates with the anchor at minimum variance.

A second question is what can be seen at all. Drift *between* historical windows is observed through tens of thousands of records, so even small changes in q_t are detectable; drift between the history and the target period is observed only through the anchor, so only large changes are. O-Drift is *observability-aware* in this sense: it segments the history where reward drift is observable, uses the anchor only for the coarse decision of which segment is closest to the target, and declines to certify an estimate when no anchor is available. The same view explains its main failure: the bias of drift-corrected reuse is too small for the anchor to observe and too large to ignore.

Our contributions are:

- a mechanism decomposition of drift for OPE, an observability analysis of which reward drift the data can reveal, and a reward-drift monitor whose size does not depend on context or logger drift (Sections 3 and 4);
- the O-Drift estimator, with segment-wise joint reward models, drift-corrected reuse, minimum-variance pooling and a random-effects interval (Section 4);
- a pre-registered evaluation on simulated regimes and semi-synthetic Open Bandit data [16], with a one-shot test split, a COPE baseline and a grouping ablation that isolates the monitor (Sections 5 and 6).

The evaluation also yields a negative result that we report in full: on Open Bandit, where the reward model is misspecified and drift is large, drift-corrected reuse is heavily biased, and two later attempts to validate reuse on the anchor or on a held-out window failed, because those samples are too small to detect the bias that matters (Section 6).

2 Related Work

Off-policy evaluation. Inverse propensity scoring [10] and the doubly robust (DR) estimator [8] are the standard tools for contextual bandit OPE; we cross-fit the reward models as in double machine learning [5]. When the data come from several loggers, weighting each record by the mixture of the logging policies (balanced or multiple importance weighting) reduces variance and stays unbiased [1,12]. Covariate shift between the logging and target populations is handled by density-ratio transport [21]. O-Drift uses all three ingredients; what it adds is the decision of which historical data the reweighted estimate may use.

Non-stationary OPE. Jagerman et al. [11] study sliding-window and exponentially decayed estimators for non-stationary bandits. Thomas et al. [20] and Chandak et al. [3,4] forecast the future value of a policy from its past per-period estimates, and Shimizu et al. [17] estimate future values by exploiting time features shared between past and future. Liu et al. [13] reuse a reward model fitted on older data inside a DR estimator on the new data. These methods weight data by time or by a model of the trend in the value; none distinguishes the mechanism behind a change.

Cross-domain OPE. COPE [14] treats other domains as sources for a target domain, clusters domains by the similarity of their mean rewards and combines multiple importance weighting inside the target cluster with a decomposed reward model. Treating time windows as domains makes COPE the closest existing estimator to ours. Its clustering statistic, the mean reward, mixes all three mechanisms; our monitor isolates the one that biases reweighted estimates. We include COPE as a tuned baseline and as a grouping ablation.

Estimator selection and drift detection. MAGIC [19], SLOPE [18] and adaptive weighting [22] choose among or blend estimators of increasing bias and decreasing variance; we use the first two as baselines over nested recency windows. Drift detection on streams is surveyed in [9]. Our monitor is a heterogeneity test [6] on per-action residuals and our interval the DerSimonian–Laird interval [7]; the novelty is the statistic tested, not the tests. Selecting segments by a preliminary test is a pre-test estimator [2], with its known risk near the threshold.

3 Problem Setting

Data. Time is divided into windows $k = 1, \dots, K$ followed by a target period T. Window k contributes logged records $\mathcal{D}_k = \{(x_i, a_i, r_i, \mu_k(a_i \mid x_i))\}_{i=1}^{n_k}$ with contexts $x \sim P_k$, actions $a \sim \mu_k(\cdot \mid x)$ from the logger deployed in window k, and binary rewards with conditional mean $q_k(x, a)$. From the target period we have a small anchor $\mathcal{D}_T$ of $n_T \ll \sum_k n_k$ logged records and a sample $\{x_j^*\}_{j=1}^{M}$ of unlabelled target contexts. The estimand for a target policy π is

$$V_T(\pi) = \mathbb{E}_{x \sim P_T}\Big[\sum_a \pi(a \mid x)\, q_T(x, a)\Big]. \tag{1}$$

Table 1. What existing estimators adapt to. Context and logger drift are handled by reweighting; "reward drift" means the method decides which data to reuse according to changes in $q_t(x, a)$.

Method	Context drift	Logger drift	Reward drift	Uses anchor	Interval
Sliding window / decay [11]	–	✓	by recency	–	–
Forecasting [20,3]	–	✓	by trend	–	✓
OPFV [17]	–	✓	by time features	–	–
Regression-assisted DR [13]	–	✓	by recency	✓	✓
SLOPE / MAGIC [18,19]	–	✓	by bias–variance	✓	–
COPE [14]	–	✓	by mean reward	✓	–
O-Drift (ours)	✓	✓	by residual test on q_t	✓	✓

Three drift mechanisms. A window can differ from the target in P_k, in μ_k or in q_k. With logged propensities and a density ratio $w_k(x) = P_T(x)/P_k(x)$, the reweighted DR score of a record in window k,

$$\Gamma_k(x, a, r) = \mathbb{E}_{x' \sim P_T}\Big[\textstyle\sum_{a'} \pi(a' \mid x')\, \hat{q}(x', a')\Big] + w_k(x)\, \frac{\pi(a|x)}{\mu_k(a|x)}\, \big(r - \hat{q}(x, a)\big), \quad (2)$$

has expectation $\mathbb{E}_{P_T}[\sum_a \pi(a \mid x) q_k(x, a)]$ for any $\hat{q}$. It equals $V_T(\pi)$ exactly when $q_k = q_T$ on the support of π under P_T. Context and logger drift therefore cost variance but not bias; reward drift costs bias that no reweighting removes. The transfer bias of window k is $b_k = \mathbb{E}_{P_T}[\sum_a \pi(a \mid x)(q_k - q_T)(x, a)]$.

Assumptions. (A1) Logged propensities are the ones used at event time. (A2) Overlap: $\pi(a \mid x) > 0$ implies $\mu_k(a \mid x) > 0$ for the windows used, and the context ratio w_k is bounded. (A3) Reward drift is piecewise: the windows split into contiguous segments with a common q within each segment. (A4) For drift-corrected reuse, the difference between segment rewards lies in the reward-model class (on the logit scale). Our regimes violate each assumption in turn (S6 breaks A1, S7 A2, S3b and S5 A3, S8 and the misspecified cells A4) so that we can see where the method fails.

Observability of reward drift. Without an anchor, b_k is not identified from history: the logs constrain only $q_1, \ldots, q_K$, any q_T is consistent with them, and no method can certify an estimate. With an anchor, b_k is identified, but only up to the noise of the anchor estimate, whose standard deviation is of order $\sigma/\sqrt{n_T}$. Biases below this level cannot be detected from the anchor, however the test is built: a level-α test with power $1 - \beta$ needs $|b_k| \gtrsim (z_{\alpha/2} + z_\beta)\, \sigma/\sqrt{n_T}$. We call drift *observable* when it exceeds this threshold for the sample that must reveal it. Drift between two historical windows is judged on n_k records and is observable down to small sizes; drift between the history and the target is judged on the n_T anchor records and is observable only when large. This asymmetry shapes the design in Section 4 and governs the failure we report in Section 6.

4 Method

O-Drift follows the observability asymmetry of Section 3: decisions that need to see small reward changes are made on history, and the anchor is used only where a coarse decision suffices. It has four steps: segment the history by reward drift, fit joint reward models, form one drift-corrected estimate per segment, and pool them with the anchor.

4.1 A reward-drift monitor

We fit one pooled per-action logistic model $\tilde{q}$ with linear and quadratic context terms on all historical windows. For a block B of windows and an action a, let $m_B(a)$ be the mean residual $r_i - \tilde{q}(x_i, a)$ over records in B with $a_i = a$, and $v_B(a)$ its binomial variance estimate. Two adjacent blocks L, R are compared by

$$\chi^2_{LR} = \sum_a \frac{\big(m_L(a) - m_R(a)\big)^2}{v_L(a) + v_R(a)}, \tag{3}$$

referred to a χ^2 distribution with one degree of freedom per action with enough records. Binary segmentation splits the history at the most significant boundary while the p-value is below α, giving contiguous segments $S_1, \ldots, S_J$ ordered from newest to oldest.

Proposition 1 (Invariance to context and logger drift). *Suppose $q_k = q$ for all windows in $L \cup R$ and $\tilde{q} \to q$. Then for every action a, $\mathbb{E}[m_L(a) - m_R(a)] \to 0$ whatever the context laws P_k and loggers μ_k, so (3) is asymptotically χ^2 and the test has level α under any context or logger drift. The mean-reward statistic $\bar{r}_L - \bar{r}_R$ and the per-action raw-reward statistic do not have this property.*

Proof (sketch). In window k, $\mathbb{E}[R - q(X, a) \mid A = a, k] = \mathbb{E}_{P_k(x|a)}[q_k(x, a) - q(x, a)] = 0$, because $P_k(x \mid a) \propto P_k(x)\mu_k(a \mid x)$ only reweights a function that is identically zero. Only $q_k \neq q$ moves $m_B(a)$. In contrast, $\mathbb{E}[R \mid k] = \sum_a \int P_k(x)\mu_k(a \mid x)q(x, a)\, dx$ changes with P_k and μ_k when q is fixed, which gives false alarms; and a reward change can be cancelled by an opposite change of context or logger, which gives missed drift. The per-action raw mean $\mathbb{E}_{P_k(x|a)}[q(x, a)]$ removes the action mix but still depends on P_k and on μ_k as a function of x.

If q lies outside the model class, the residual has conditional mean $\mathbb{E}_{P_k(x|a)}[q - \tilde{q}]$, which depends on P_k; context drift can then trigger the monitor. The richer the model class, the smaller this effect.

4.2 Joint segment reward models

Reward models for estimation are cross-fitted [5] per-action logistic regressions on a context basis $\phi(x)$. Across segments we fit one joint model

$$\operatorname{logit} \hat{q}_j(x, a) = \phi(x)^\top \beta_a + \phi(x)^\top \delta_{j,a}, \qquad \delta_{1,a} = 0, \tag{4}$$

a shared base plus per-segment offsets, with an L_2 penalty on the offsets scaled by $1/s^2$. The scale $s \leq 1$ acts as a ridge prior on drift size and interpolates between fully corrected ($s = 1$) and uncorrected reuse ($s \to 0$). The basis is chosen per episode between linear features and $D = 32$ random Fourier features [15] by held-out log-loss of the pooled model; the same basis is used by every method.

4.3 Reference segment and drift-corrected reuse

The reference segment is the newest segment whose uncorrected estimate the anchor does not contradict at $|z| \leq z_{\text{anc}}$; if all are contradicted the anchor is used alone. Each segment S_j contributes

$$\hat{V}_j = \frac{1}{M} \sum_{m=1}^{M} \sum_{a} \pi(a \mid x_m^*) \hat{q}_{\text{ref}}(x_m^*, a) + \frac{1}{n_j} \sum_{i \in S_j} W_i \left(r_i - \hat{q}_j(x_i, a_i) \right), \tag{5}$$

$$W_i = \frac{\pi(a_i \mid x_i)}{\sum_{k \in S_j} \frac{n_k}{n_j} \mu_k(a_i \mid x_i) / w_k(x_i)}. \tag{6}$$

The direct-method term uses the reference model, so it targets q_{ref}; the correction term uses the segment's own model, so it removes the segment's own model error. W_i is the balanced importance weight from the target to the segment's mixture of loggers and context laws [1,12], with w_k estimated by a probabilistic classifier. For the reference segment $\hat{q}_j = \hat{q}_{\text{ref}}$ and (5) is the usual balanced DR estimate. For an older segment, the difference $\hat{q}_{\text{ref}} - \hat{q}_j$ is exactly the fitted reward drift, so (5) transfers the segment's evidence to the reference reward when (A4) holds.

4.4 Pooling and interval

The anchor DR estimate joins the segment estimates, and the weights $\hat{\alpha} = \arg\min_{\alpha \geq 0, \mathbf{1}^\top \alpha = 1} \alpha^\top \hat{\Sigma} \alpha$ minimise the estimated variance, where $\hat{\Sigma}$ accounts for the target contexts shared by all direct-method terms. Because bias may remain, the reported interval is DerSimonian–Laird [7], inflating the standard error by the between-component heterogeneity $\hat{\tau}^2$; a sample-split interval with an explicit bias allowance is reported as a certified but wide alternative. Without an anchor O-Drift returns an estimate flagged as not certified.

Settings. $\alpha = 10^{-4}$, $z_{\text{anc}} = 5$ and $s = 0.25$ were tuned on development data (Section 5).

5 Experimental Setup

Simulated regimes. Each environment has $d = 5$ Gaussian contexts, 10 actions and per-action logistic rewards with a linear part and a nonlinear part of weight 0.5 ("misspecification 0.5") or 3.0. The level 3.0 was fixed by matching the nonlinear share of logit variance in Open Bandit (0.43 vs. 0.41) before any result

at that level was seen. The history has $K = 12$ windows of 5,000 records; the anchor has 1,000 records and there are 10,000 target contexts. Ground truth is exact over actions and Monte Carlo over 10^6 contexts. The regimes isolate one mechanism at a time: S0 stationary; S1 context drift growing with age; S2 logger switches A→B→C; S3a abrupt reward drift in the older half; S3b gradual reward drift towards the target; S4 mixed (oldest third reward-drifted, middle third context-shifted, newest third poorly explored); S5 seasonal context and reward drift; S6 wrong logged propensities (violates A1); S7 recent windows that never play three actions (violates A2); S8 reward drift along an out-of-class interaction, S8b the same without an in-class component (violate A4). Intensities are given with each regime in Table 2. Four candidate policies of increasing noise around the optimal policy are evaluated in every episode.

Open Bandit semi-synthetic data. We use the uniformly random logging data of the Men campaign of the Open Bandit Dataset [16] (452,949 records, 34 items). Contexts are the real user features reduced to 8 principal components, in their real temporal order, split into 13 blocks of equal size (12 history windows and the target block). The base reward is a per-action logistic model with linear and quadratic terms fitted to the real clicks and shifted to a mean click rate of 0.05; actions, propensities, rewards and the injected logger and reward drift are synthetic. Because the generating model is quadratic, the methods' reward models are misspecified. These data test robustness, not natural drift.

Baselines. All share the reward models and basis rule: pooled, anchor-only, recent-k and decayed DR [11]; minimum-variance pooling; transport DR and its recent, decayed and balanced-decayed variants; regression-assisted DR [13]; SLOPE [18] and MAGIC [19] over nested windows; forecasting [20]; time-aware DR (polynomial time trend in the reward model); balanced DR [1]; an anchor-validated window selector; COPE [14] (windows as domains, anchor as target, cluster size tuned); and monitor-select, DR on the newest monitor segment plus the anchor without reuse.

Protocol. The protocol (`configs/protocol.yaml` in the code) was fixed before tuning. Every method family, baselines included, is tuned on a development split (four environments, 200 episodes per cell) by minimising the mean over cells of log RMSE. Test splits use unseen environments (300 episodes per cell; 200 for Open Bandit) and are run once per frozen configuration. The version reported here is the third pre-registered one: the first passed its simulator criteria on separate test seeds but failed on Open Bandit, which led to the misspecification-3 cells, the robustness criteria, the per-episode basis, offset shrinkage, the COPE baseline and the grouping ablation, all registered before this test run. Twelve criteria were pre-registered: five on accuracy, coverage, support and abstention (C1–C5), two on robustness in the misspecification-3 and Open Bandit cells (R1: no ratio above 1.5; R2: ratio at most 1 in at least half of them), two against COPE (N1) and three on the grouping ablation (N2); their exact wording is in the protocol file.

Table 2. One-shot test results on the simulated regimes (300 episodes × 4 policies per cell). RMSE ($\times 10^{-3}$) of O-DRIFT, of its configuration without offset shrinkage (v1) and of the best tuned baseline in the cell; ratio of O-DRIFT to that baseline and coverage of its nominal 95% interval. Mechanism checks: RMSE ratio of O-DRIFT to tuned COPE and of the blind-grouping ablation (mean-reward statistic) to O-DRIFT; share of episodes split into more than one segment by the residual monitor (R) and by the blind statistic (B). *: paired 95% interval of the RMSE difference excludes 0.

			Accuracy						Mechanism checks			
Reg.	Int.	Mis.	O-DRIFT	v1	Best baseline	RMSE	Ratio	Cov.	COPE	Blind	R	B
S0	0	0.5	5.4	5.4	Min-var DR	5.1	1.05*	0.93	1.00	1.00	0.00	0.00
S1	0.5	0.5	5.5	5.5	COPE	5.5	1.00	0.95	1.00	1.07*	0.00	0.22
S1	2	0.5	8.0	8.0	COPE	7.8	1.02	0.93	1.02	1.07*	0.00	0.61
S2	0	0.5	5.1	5.1	Balanced DR	4.7	1.07*	0.95	1.01	1.13*	0.00	0.52
S3a	0.25	0.5	6.4	6.5	Monitor-select DR	7.0	0.93*	0.89	0.52*	1.56*	1.00	0.29
S3a	1	0.5	9.8	6.5	Monitor-select DR	7.0	1.40*	0.83	0.25*	1.03	1.00	0.99
S3b	0.25	0.5	10.9	10.8	Time-DR	8.5	1.28*	0.63	0.82*	1.21*	0.64	0.02
S3b	1	0.5	26.6	18.4	Time-DR	8.8	3.02*	0.27	0.60*	1.14*	1.00	0.82
S4	0.25	0.5	7.5	8.0	Balanced DR	8.6	0.86*	0.95	0.62*	1.25*	1.00	0.88
S4	1	0.5	14.1	14.8	T-recent DR	21.7	0.65*	0.78	0.32*	1.02	1.00	1.00
S5	0.5	0.5	21.5	20.9	Decay DR	20.4	1.05*	0.34	0.93*	1.06*	0.99	0.72
S5	1	0.5	34.7	28.9	SLOPE	36.0	0.96	0.22	0.84*	1.13*	1.00	0.90
S6	0	0.5	5.9	5.9	Balanced DR	5.3	1.13*	0.94	1.07*	1.08*	0.00	0.52
S7	0	0.5	7.4	7.4	Min-var DR	4.9	1.52*	0.94	1.30*	1.02*	0.00	1.00
S8	0.5	0.5	6.7	6.7	Monitor-select DR	7.0	0.96	0.88	0.42*	1.58*	1.00	0.51
S8	1	0.5	6.9	6.8	Monitor-select DR	7.0	0.98	0.86	0.24*	2.24*	1.00	0.50
S8b	1	0.5	6.7	6.7	Monitor-select DR	7.0	0.96	0.89	0.49*	1.23*	1.00	0.65
S0	0	3	4.9	4.9	Pooled DR	4.5	1.08*	0.95	1.00	1.00	0.00	0.00
S2	0	3	5.0	5.0	Balanced DR	4.6	1.08*	0.95	1.00	1.26*	0.00	0.59
S3a	1	3	6.5	6.7	Monitor-select DR	6.7	0.98	0.86	0.19*	2.35*	1.00	0.72
S4	1	3	17.9	25.7	TB-decay DR	22.0	0.81*	0.70	0.48*	1.03*	1.00	0.99
S5	1	3	23.3	26.2	Monitor-select DR	29.2	0.80*	0.33	0.66*	1.28*	1.00	0.72
S8	1	3	6.6	6.7	Monitor-select DR	6.7	0.99	0.86	0.22*	1.01	1.00	1.00

Metrics. RMSE ($\times 10^{-3}$); its ratio to the best tuned baseline with a paired bootstrap 95% interval of the difference (10,000 resamples); interval coverage; and the share of episodes split.

6 Results

All numbers in this section come from the one-shot test runs.

Simulated regimes. Table 2 compares O-DRIFT with the best tuned baseline in each cell, which differs from cell to cell. Without reward drift (S0–S2) O-DRIFT is within 7% of the best baseline; the small losses in S0 and S2 are the price of estimating segment models that turn out to be unnecessary. Under abrupt and mixed reward drift it wins: by 7% in S3a at the lower intensity and by 14% and 35% in S4, with paired intervals that exclude zero. In S4 the best baseline changes from balanced DR to transported recent-window DR as the intensity grows, while O-DRIFT is best in both. The misspecification-3 cells of S3a, S4, S5 and S8 are also won or tied (ratios 0.80–0.99).

It loses in four places. Gradual drift (S3b) violates the piecewise assumption and favours the trend-modelling time-aware DR (1.28, 3.02). Under abrupt

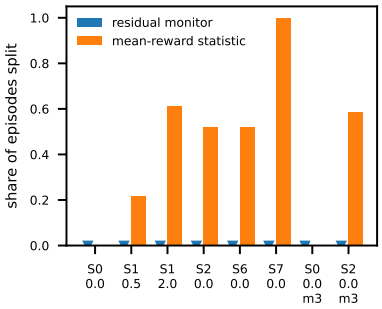


Fig. 1. Share of test episodes in which the history is split into more than one segment, in regimes without reward drift, for the residual monitor (zero in every cell) and for the mean-reward statistic (same segmentation procedure and threshold grid).

drift at the higher intensity (S3a, intensity 1) O-Drift loses to monitor-select (1.40); here the v1 configuration without offset shrinkage would have won (6.5 vs. 7.0), so the shrinkage tuned on development data under-corrects large drift. The same holds for S5. In S7 (support loss) the estimator correctly puts no weight on support-deficient windows, but loses accuracy against min-variance pooling (1.52). The heterogeneity-inflated interval undercovers whenever bias remains, most severely in S5 and S3b (0.22–0.63); the certified sample-split interval covers in every regime but is about 18 times wider.

Does the mechanism matter? Two pre-registered checks address this (right half of Table 2, Figure 1). Against tuned COPE, which groups windows by mean reward, O-Drift has lower RMSE in all 15 reward-drift cells (ratios 0.19–0.93, intervals excluding zero) and ties it in S0–S2; COPE is better only in S6 and S7 (1.07, 1.30), the regimes with faulty propensities or lost support. Tuned COPE keeps all windows in its cluster, so grouping by mean reward did not beat pooling. The grouping ablation keeps the whole O-Drift pipeline and replaces only the monitor statistic by the mean reward. The residual monitor never splits the history when the reward is stable (S0–S2, and S6–S7), as Proposition 1 predicts, whereas the mean-reward statistic splits 22–61% of S1 histories under context drift, 52% of S2 histories under logger switches and every S7 history. These spurious splits shrink the reference segment and cost 7–13% RMSE in S1 and S2. Under reward drift the blind grouping is never better and is worse in most cells (up to 2.35×), mainly because it misses abrupt changes: in S3a at the lower intensity it detects the change in 29% of episodes, against 100% for the monitor.

Open Bandit: where drift-corrected reuse fails. Table 3 shows the pre-registered robustness test on Open Bandit. Without reward drift O-Drift matches the best baseline, but under reward drift it is 3.1–7.7 times worse, and almost all of its error is bias ($+14$ to $+68\times10^{-3}$). The monitor is not the cause: it finds the change in every S3a episode, and monitor-select, which uses the same segmentation without reuse, is the best method in S3a and within a factor of 1.4 of the best in S4. The failure lies in the correction. The base reward is quadratic, the reward models linear (the basis rule chose linear features in every episode), and the

Table 3. One-shot test on semi-synthetic Open Bandit data (200 episodes per cell). RMSE and mean error (bias) $\times 10^{-3}$; v1 is O-DRIFT without offset shrinkage; monitor-select uses the same segmentation without reuse. Ratio as in Table 2.

Regime	Int.	O-DRIFT	Bias	v1	Monitor-select	Best tuned baseline	RMSE	Ratio
S0	0	3.2	+0.4	3.2	3.3	COPE	3.1	1.03
S2	0	2.8	+0.3	2.8	3.6	COPE	2.8	1.00
S3a	0.5	15.7	+13.6	7.5	4.7	Monitor-select DR	4.7	3.33*
S3a	1	36.4	+32.2	10.0	4.7	Monitor-select DR	4.7	7.74*
S4	0.5	32.0	+28.1	21.1	13.6	Time-DR	10.2	3.14*
S4	1	73.8	+66.4	31.8	13.5	T-recent DR	10.7	6.87*
S5	1	75.4	+68.0	58.5	27.3	Reg.-DR (Liu)	20.7	3.63*

Table 4. Post-hoc revisions on development data (52 matched episodes per cell). RMSE of each version divided by the RMSE of the best tuned baseline (shown, $\times 10^{-3}$). v1: no offset shrinkage; v3: shrinkage 0.25 (the tested version); v4: reuse gate at $z = 1.96$; v5: rolling-origin back-test, with the share of episodes in which it chose full reuse, shrunk reuse or no reuse.

				RMSE ratio				v5 choice		
Reg.	Int.	Mis.	Best	v1	v3	v4	v5	Reuse	Shrink	None
S0	0	0.5	6.3	1.00	1.00	1.00	1.01	0.50	0.00	0.50
S1	1	0.5	7.0	0.96	0.96	0.96	1.42	0.69	0.00	0.31
S2	0	0.5	5.5	0.99	0.99	0.99	1.23	0.58	0.00	0.42
S3a	0.5	0.5	7.7	0.89	0.86	0.89	0.95	0.33	0.27	0.40
S3b	0.5	0.5	8.5	1.02	1.09	1.02	0.99	0.31	0.27	0.42
S4	0.5	0.5	11.9	0.89	0.71	1.32	1.53	0.29	0.38	0.33
S0	0	3	5.1	1.00	1.00	1.00	0.99	0.46	0.00	0.54
S3a	0.5	3	7.1	0.93	0.85	0.94	0.97	0.27	0.33	0.40
S4	0.5	3	10.9	1.32	0.94	1.61	1.79	0.35	0.31	0.35

drift three times the standard deviation of the base logit, so the offsets δ_j absorb model error as well as drift and the corrected segments carry a bias the anchor cannot observe. Offset shrinkage makes this worse (S3a, intensity 1: 36.4 vs. 10.0 without shrinkage). The robustness criteria R1 and R2 therefore fail. Of the twelve pre-registered criteria, the other ten pass (C1–C5 and N1–N2).

Can reuse be validated on the data at hand? (post-hoc). After this test we tried, on development data only, two ways to decide per episode whether to reuse corrected segments (Table 4). A gate (v4) that drops a segment whose estimate differs from the reference (reference segment plus anchor) at the 5% level removes the bias but raises the variance, and loses in S4 (1.32× the best baseline) and in S4 at misspecification 3 (1.61×). A rolling-origin back-test (v5) that treats the newest history window as a pseudo-target and chooses among full reuse, shrunk reuse and no reuse. It picks each option in roughly a third of the episodes in every reward-drift cell, and no reuse in 31–54% of the episodes without reward drift, where reuse is exactly right; it is worse than the best fixed version in six of nine cells (up to 2.1×) and tied in a seventh. Neither was taken to a test split. Both fail for the reason given in Section 3: an anchor of 1,000 records gives a DR estimate with RMSE 42×10^{-3}, and a held-out window of 4,000 records

about half of that, while the reuse bias that separates good from bad decisions is around 10×10^{-3}.

7 Limitations and Conclusion

For OPE under drift, which mechanism changed a window matters more than its age, and a drift decision is only as good as the data that can observe it. A residual test on the conditional reward does not mistake context or logger drift for reward drift, which in our pre-registered evaluation separated O-Drift from mean-reward grouping in every reward-drift setting.

The limitations are as important as the result. First, drift-corrected reuse depends on the reward-drift difference lying in the model class (A4); on semi-synthetic Open Bandit data, with a misspecified model and large drift, it was heavily biased, and the estimator without reuse was the better choice. Second, data-driven checks of reuse on the anchor or on a held-out window could not repair this, because samples of the size available per period cannot detect the biases involved; in our terms, the reuse bias is not observable at the anchor's sample size. We see this as an information limit, not a tuning problem, and we expect it to bind any method that validates reuse on the current period alone. Third, gradual drift violates the piecewise assumption and favours methods that model a trend. Fourth, our heterogeneity-adjusted interval undercovers when bias remains, and the certified alternative is wide. Finally, Open Bandit provides real contexts and their real order but synthetic rewards and drift; an evaluation on naturally drifting logged data remains open.

Two directions follow: reward models flexible enough to make (A4) plausible, so that reuse is safe when attempted, and reporting for each estimate how much unobservable drift would change a deployment decision.

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